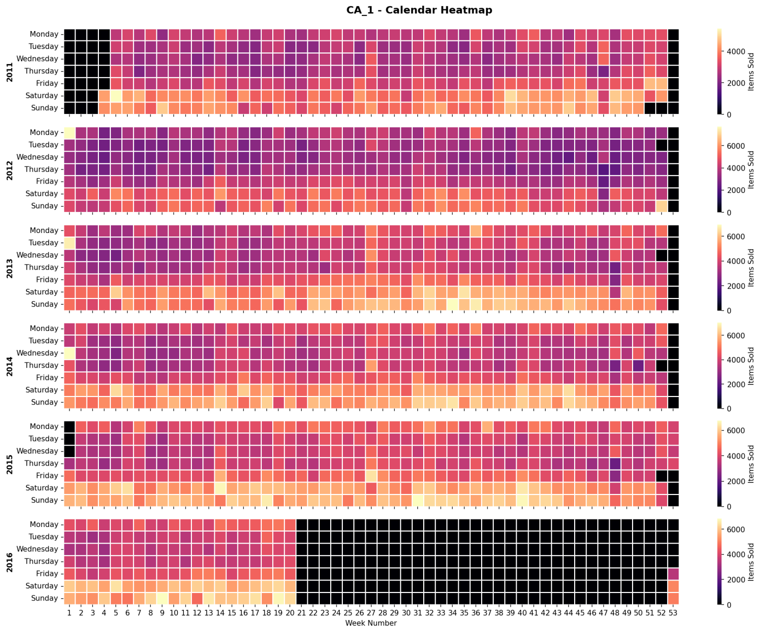
MATH 5470 Mini-Project 1: M5 Forecasting – Accuracy

Won Joon Kim   wjkimab@connect.ust.hk
Department of Chemical and Biological Engineering, School of Engineering, The Hong Kong University of Science and Technology

1. Introduction to the Dataset and Aim

The M5 dataset (Walmart) involves the unit sales of various products sold in the USA, organized in the form of grouped **time series**. More specifically, the dataset involves the unit sales of 3,049 products, classified in 3 product categories (Hobbies, Foods, and Household) and 7 product departments. The products are sold across **10 stores**, located in 3 states (CA, TX, and WI). This creates a dataset which is **hierarchical** in nature. The aim is to accurately estimate the unit sales of Walmart retail goods in the next 28 days.

2. Exploratory Data Analysis



We can see from the heatmap that for the first store in California, there is a general increase in items sold towards the weekends. The black squares denote missing data, which is intrinsic to this dataset. Although subtle, there is a gradual increase in sales over time (i.e., less purple). From the violin plot, we can see that household items on median cost the most, followed by hobbies and food. This trend is observed in all stores across the USA.

3. Feature Engineering for Time Series Modeling

From reviewing public Kaggle notebooks, we noted that tree-based boosting methods (i.e., LightGBM and XGBoost) are the most performative and popular for this dataset. It is likely that this stems from the hierarchical data structure, which tree-based methods can effectively capture.

Time series data must be re-framed as a supervised learning dataset before we can start using machine learning algorithms. There is no concept of input and output features. Instead, we must choose the variable to be predicted and use feature engineering. The goal of feature engineering is to provide strong and ideally simple relationships between new input features and the output feature for the supervised learning algorithm to model.

To reduce memory usage, we used a custom "downcasting" function to simplify data formats (e.g., categorical and less floating points). We "melted" the sales into long format (i.e., 1,969 rows for each item sold), and merged with calendar and prices, which ended up as a large dataframe of over 6 million rows. We added "lag" features starting from 29 to ensure we never use future information, and ending at 60 to allow long term trends to be captured. We added "new" features by taking the mean and median of the "state", "store", "item_id", "item_state", and "item_store" columns.

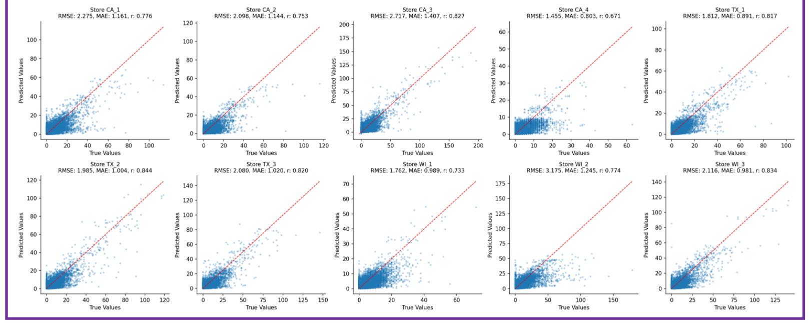
4. Discussion and Future Work

We note that there may have been evaluation data leakage during the training process, however, due to the time constraint, it was not properly addressed. This may lead to an overestimated model performance.

5. Optimized Light Gradient Boosting Machine

We trained 10 separate models, 1 for each store, as the sales patterns are widely dissimilar. The hyperparameters for lightgbm was determined after a tuning run as follows: n_estimators=200, learning_rate=0.1, num_leaves=50, and we employed model regularization via early_stopping=20.

The final model performance was measured with the root mean square error metric, and was **2.116**. As this value is context-dependent, we further quantified via the Pearson correlation coefficient between the predicted and actual target values, and observed a satisfactory $r > 0.65$ for all models.



6. Final Kaggle Competition Submission

Submission and Description	Private Score	Public Score
submission.csv Complete (after deadline) · 20m ago	0.77488	0.87742

7. References

Addison Howard, inversion, Spyros Makridakis, and vangelis. M5 Forecasting - Accuracy. https://kaggle.com/competitions/m5-forecasting-accuracy, 2020. Kaggle.
Anshul Sharma. Time Series Forecasting-EDA, FE & Modelling. https://www.kaggle.com/code/anshuls235/time-series-forecasting-eda-fe-modelling/notebook, 2020. Kaggle.
KevinFFF. math5470_Fan-M5accuracy. https://www.kaggle.com/code/kevinfff/math5470-fan-m5accuracy, 2025. Kaggle.

8. Acknowledgement and Code Availability

Won Joon Kim conducted review of existing notebooks, exploratory data analysis, feature engineering, model training, and model prediction. Source code can be found at https://github.com/wjoonkim/math5470_Kim.